ToddySoft Connect ADS Driver - Ignition 8.1

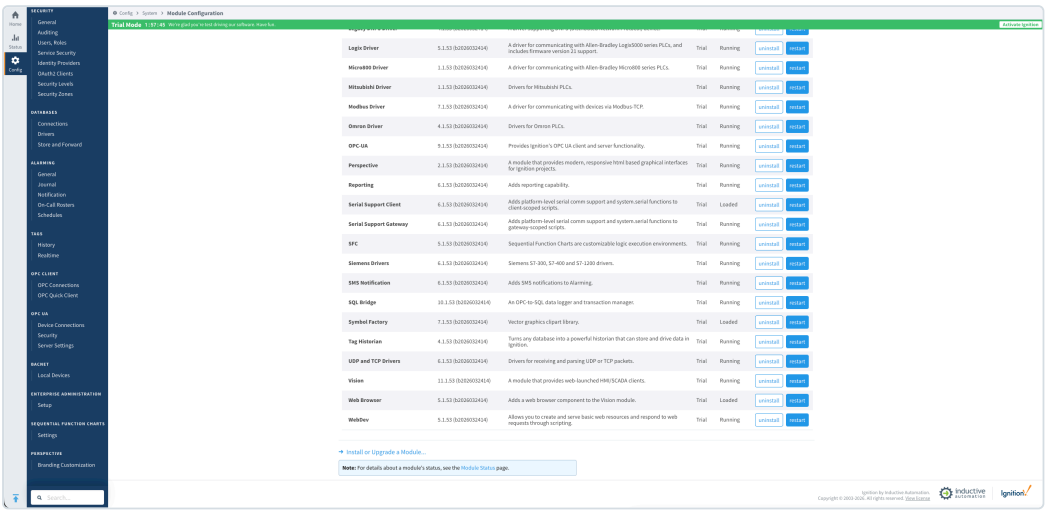
Ignition 8.1 module for connecting to Beckhoff TwinCAT PLCs via the ADS/AMS protocol, with support for plain TCP and encrypted "Secure ADS" (TLS) connections.

Installation

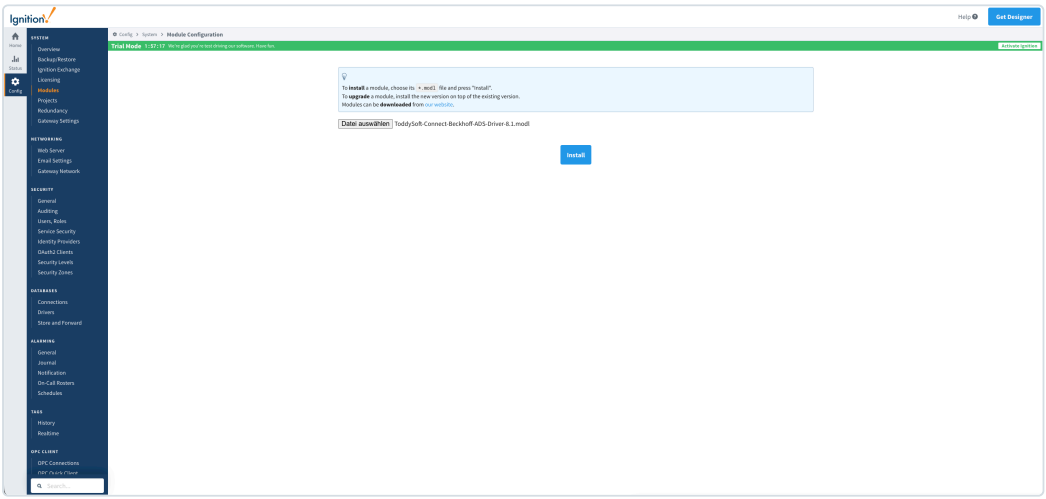
> **This module is for Ignition 8.1 only.** It is built against the Ignition 8.1 SDK and will not run on any other major version. For Ignition 8.3, install the separate `…-8.3.modl` build instead — the two are not interchangeable, and each gateway needs the build that matches it.
>
> **Ignition 8.0 and earlier are not supported and cannot be.** The module is built on the device-driver API that Ignition introduced with 8.1, so on 8.0 the gateway fails to load it with `NoClassDefFoundError: … opcua.server.api.AbstractDeviceModuleHook`.
>
> Both builds carry the same name in the gateway's module list, so tell them apart by the file you installed or by the module description, which names the Ignition version.
>
> **Ignition's OPC UA module must be installed and running.** This is an OPC UA device driver: it registers with Ignition's own OPC UA server, and every tag it exposes is served through it. With that module absent the gateway cannot load this one, and the log fills with `ClassNotFoundException` / `NoClassDefFoundError` naming `com.inductiveautomation.opcua…` classes. The OPC UA module ships with Ignition and is installed by default, so in practice this bites only on a gateway where it has been removed or disabled. **Ignition 8.1 cannot refuse the module for you** — its descriptor has no way to mark a dependency as required, a mechanism that arrived in 8.3 — but it does name the cause plainly, one line before the stack trace. In `logs/wrapper.log`, look for:
>
> `Module "com.toddysoft.connect.…" depends on "com.inductiveautomation.opcua", but it doesn't exist.`
>
> It then attempts to load the module anyway, which produces the class errors above. Ignition's own drivers behave identically in this situation, so this is how the platform works and not a fault in this module.

The ToddySoft Connect modules are signed with a self-signed certificate (issued to *ToddySoft GmbH*) rather than a publicly accredited code-signing certificate. Ignition therefore asks you to review and accept that certificate once during installation - it is then remembered for all future ToddySoft modules signed with the same certificate.

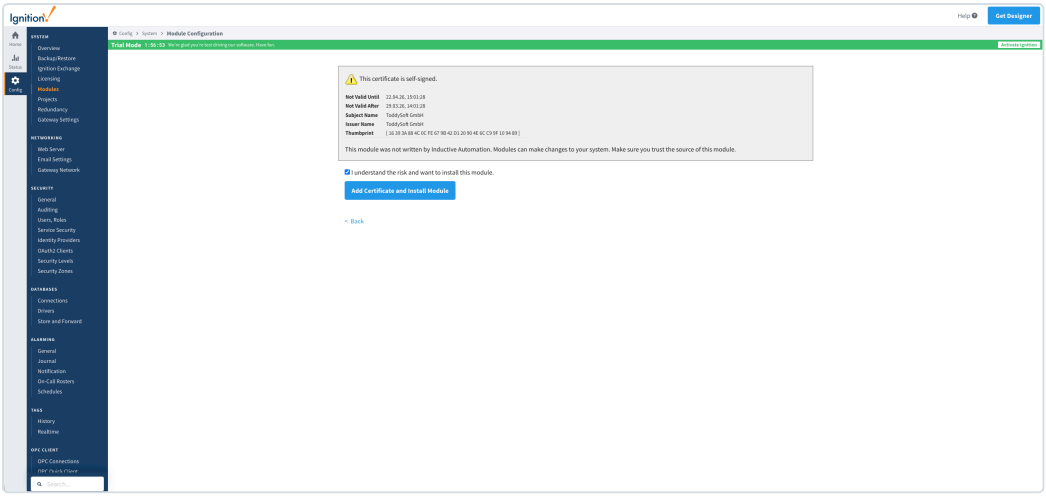
1. Open the Ignition Gateway web interface and navigate to **Config > System > Modules**. Scroll to the bottom and click **Install or Upgrade a Module...**.



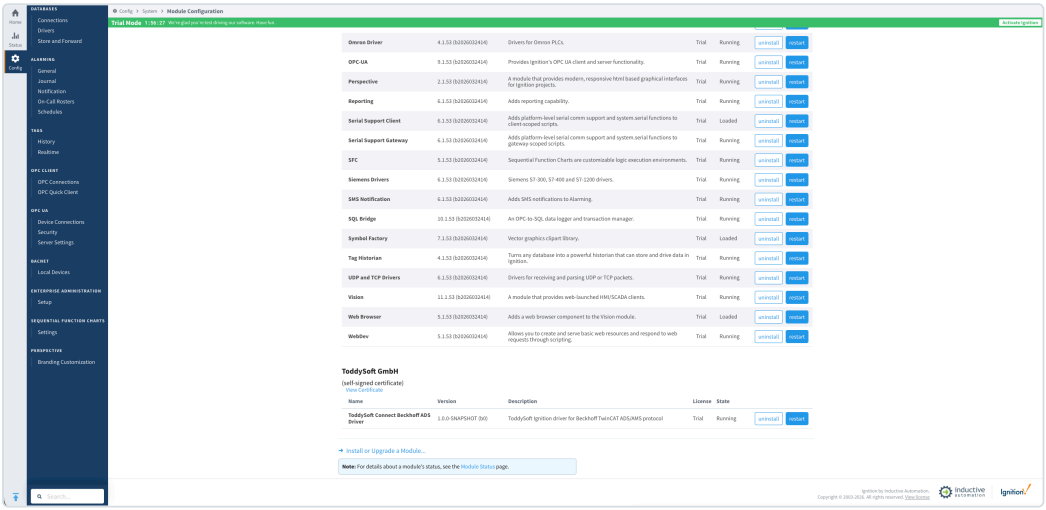
2. Choose the `ToddySoft-Connect-Beckhoff-ADS-Driver-8.1.modl` file and click **Install**.



3. Ignition shows the self-signed certificate details (subject and issuer *ToddySoft GmbH*) together with a warning that the module was not written by Inductive Automation. Review the certificate, tick **I understand the risk and want to install this module** and click **Add Certificate and Install Module**.

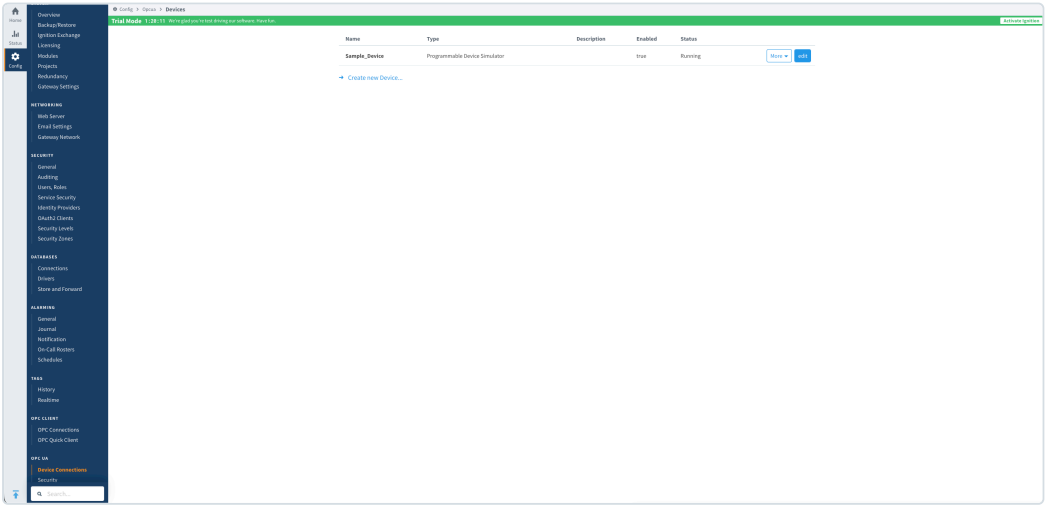


4. The module is now trusted and listed under the **ToddySoft GmbH (self-signed certificate)** group with the state **Running**.

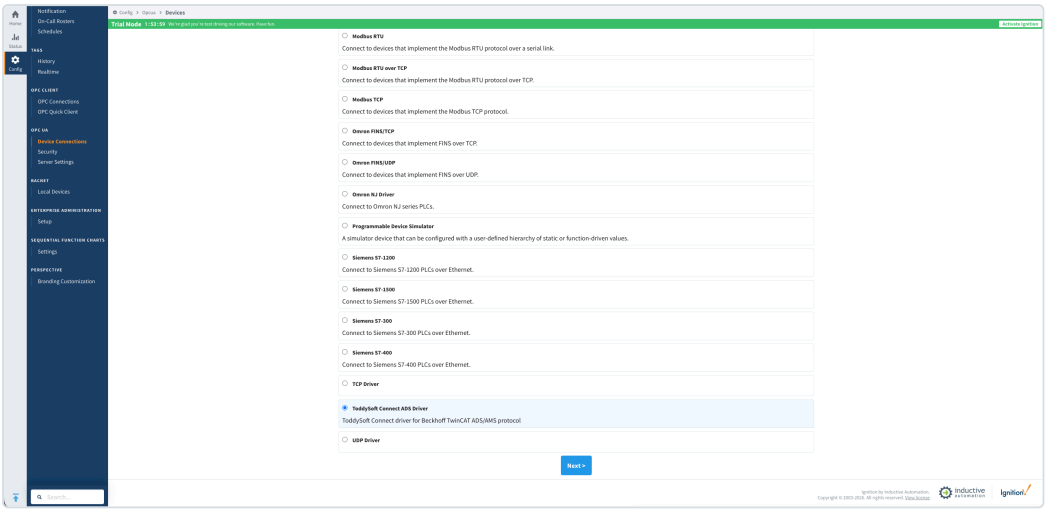


Creating a Device Connection

1. Go to **Config > OPC UA > Device Connections**.

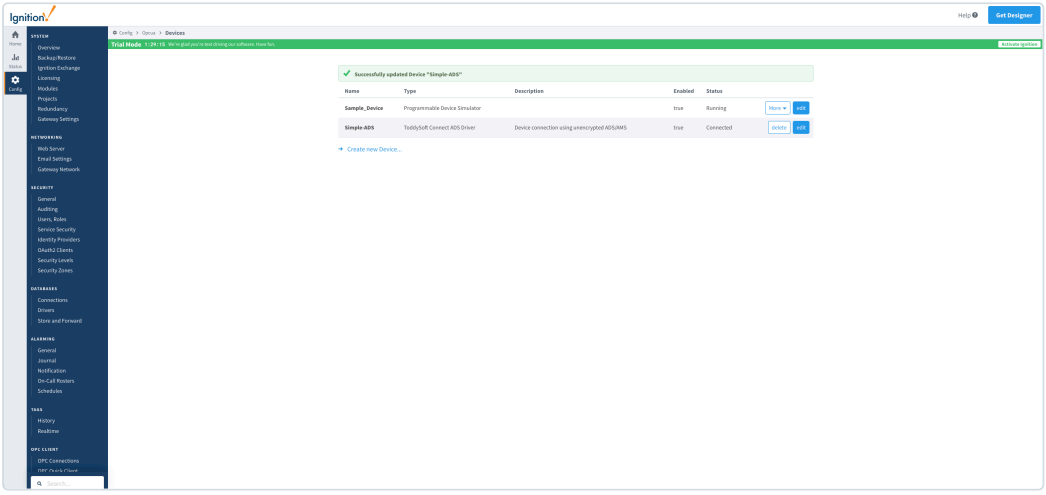


2. Click **Create new Device...**, select **ToddySoft Connect ADS Driver** as the device type and click **Next**.



3. Configure the connection settings (see the sections below) and click **Save**.

4. The new device then appears in the Device Connections list.

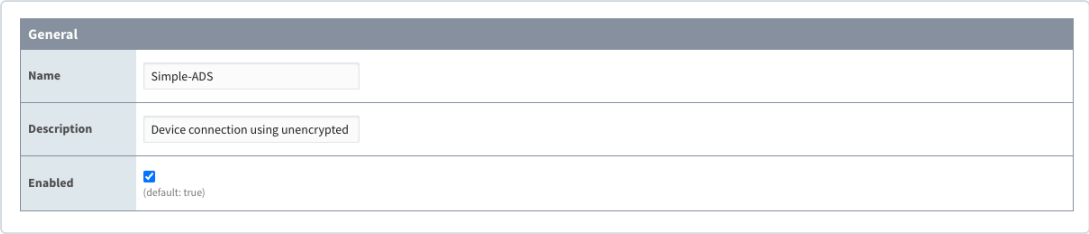


Setting blocks

The configuration is organized into categories: **General**, **Transport Settings**, **ADS Settings**, **TCP Advanced Settings**, **Audit Log Settings** and **TLS Settings**.

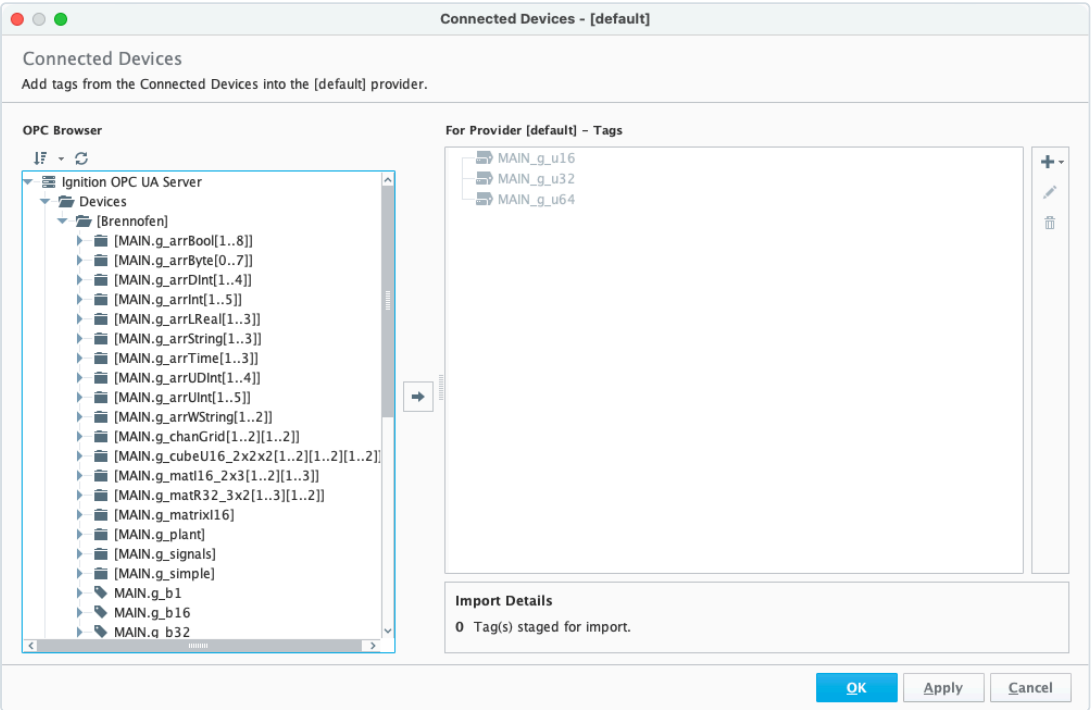
General

This block generally simply assigns a name to the connection.



Setting	Description	Default
Name	Unique name for this device connection	
Description	Optional description for this connection	
Enabled	Whether the connection is active	`true`

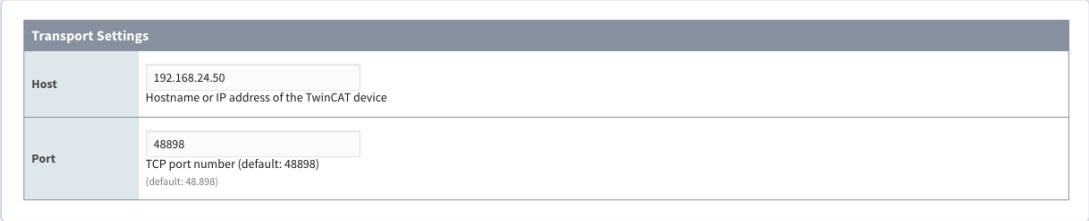
This is the name the connection is then referenced under in other parts of Ignition:



If `Enabled` is not checked, no connection is established until it is enabled.

Transport Settings

The endpoint of the TwinCAT device.



Setting	Description	Default
Host	Hostname or IP address of the TwinCAT device	`localhost`
Port	TCP port number (default: 48898, TLS: 8016)	`48898`

ADS Settings

This block configures the ADS-specific settings.



Setting	Description	Default
Target AMS Net ID	AMS Net ID of the target device (format: `x.x.x.x.x.x`). Defaults to `{target-ip}.1.1`	*(auto)*
Target AMS Port	AMS port of the target device	`851`
Source AMS Net ID	AMS Net ID of the source (format: `x.x.x.x.x.x`). Defaults to `{client-ip}.1.1`	*(auto)*
Source AMS Port	AMS port of the source (typically 30000-65535). Defaults to the locally assigned TCP port	*(auto)*
Request Timeout	Default timeout for all types of requests in milliseconds	`4000`
Browse Queries	Query string for symbol browsing. Multiple values separated with `;`	`MAIN.*`

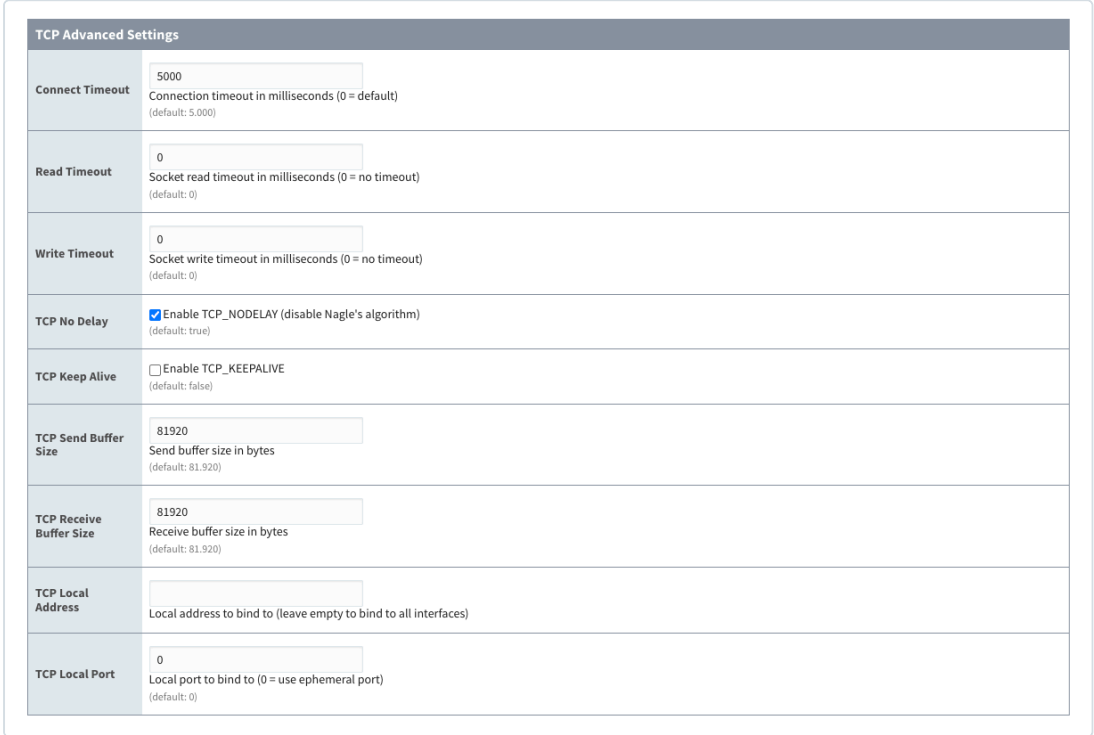
If not provided in the `Target AMS Net ID` field, the value is calculated by appending `.1.1` after the local IP address of the TCP connection. Please beware that there could be multiple paths to reach the PLC (if your device is assigned multiple IP addresses). If you want to control the IP address used, please refer to the `TCP Local Address` setting in the `TCP Advanced Settings` section.

Common AMS ports:

Port	Description
851	TwinCAT 3 PLC Runtime 1 (default)
852	TwinCAT 3 PLC Runtime 2
853	TwinCAT 3 PLC Runtime 3
801	TwinCAT 2 PLC Runtime 1
811	TwinCAT 2 PLC Runtime 2

> **Note (Source AMS Net ID and PLC routes):** For plain TCP connections, the target PLC only accepts ADS from AMS Net IDs it already has a route for. If the connection is refused immediately (the PLC closes the TCP connection), add a static route on the TwinCAT side for the source AMS Net ID shown in the gateway log. The TLS modes below create this route automatically.

TCP Advanced Settings



Setting	Description	Default
Connect Timeout	Connection timeout in milliseconds (0 = default)	`5000`
Read Timeout	Socket read timeout in milliseconds (0 = no timeout)	`0`

Setting	Description	Default
Write Timeout	Socket write timeout in milliseconds (0 = no timeout)	`0`
TCP No Delay	Enable TCP_NODELAY (disable Nagle's algorithm)	`true`
TCP Keep Alive	Enable TCP_KEEPALIVE	`false`
TCP Send Buffer Size	Send buffer size in bytes	`81920`
TCP Receive Buffer Size	Receive buffer size in bytes	`81920`
TCP Local Address	Local address to bind to (leave empty to bind to all interfaces)	*(empty)*
TCP Local Port	Local port to bind to (0 = use ephemeral port)	`0`

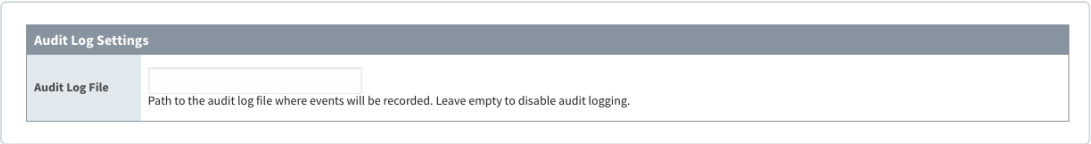
Limits

Ceilings on what the controller can make the driver allocate. Each one bounds something the *other* side chooses - a declared table depth, a nesting depth, the size of a sum command - so a device that misbehaves cannot exhaust the gateway. The defaults are the driver's own; raise one only for a controller measured to need it. Leaving them alone produces exactly the connection the device made before these settings existed.

Setting	Description	Default
Max Browse Depth	How deep the browser dereferences a symbol into child nodes. A type declared in terms of another that refers back to it forms a cycle that parses fine and then walks forever.	`20`
Max Data-Type Table Depth	Nesting depth accepted when parsing the data-type table uploaded from the device. Bounds how deeply the table is *parsed*, where Max Browse Depth bounds how deeply it is *dereferenced*.	`20`
Max Sum Sub-Commands	Sub-commands packed into one Sum-Read or Sum-Write; a larger request is split across several. The default is Beckhoff's own recommendation - the PLC collects and copies every variable in the request before it can start its next cycle.	`500`
Max Sum Bytes	Bytes one Sum-Read or Sum-Write may move through the AMS router; a larger request is split. The default is the AMS router's own limit.	`2097152`

Audit Log Settings

The `Audit Log` logs important information to a dedicated log file. This will grow in size very quickly, so it is advised to only use this setting when investigating issues.



Setting	Description	Default
Audit Log File	Absolute path to the audit log file where events will be recorded. Leave empty to disable.	*(empty)*

NOTE: Please ensure the user the `Ignition` process is running under has write access to this path.

The Audit Log system will automatically create the file and create compressed files next to it, if a certain size is reached or on a time-based schedule.

TLS Settings

By default the ADS connection uses plain, unencrypted TCP. To encrypt the communication ("Secure ADS"), enable TLS and select one of the three supported authentication modes. The two settings shared by all encrypted modes are:

Setting	Description	Default
TLS Enabled	Encrypt the ADS connection using TLS	`false`
TLS Authentication Mode	Authentication method: `Pre-Shared Key`, `Self Signed Certificate` or `Shared Certificate Authority`	`Self Signed Certificate`

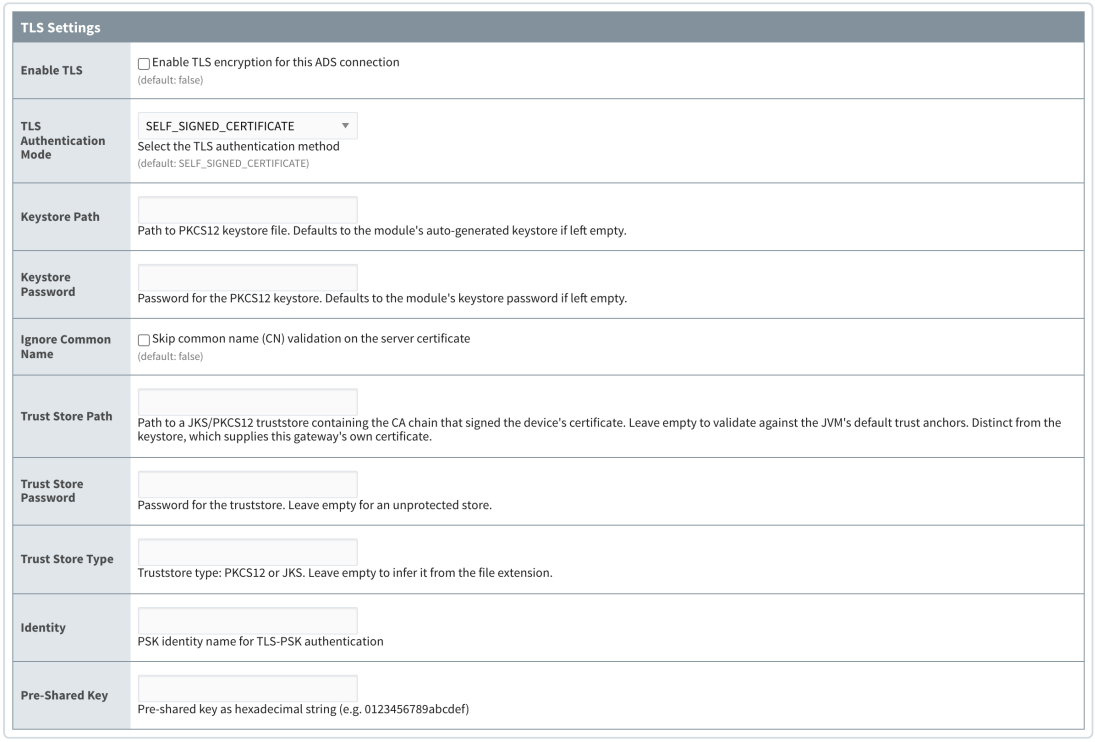
When TLS is enabled, the connection port automatically switches from `48898` to `8016`, unless a non-default port has been set explicitly in the Transport Settings.

> **Storing passwords and keys:** The password and key fields used by the encrypted modes below (`Keystore Password` and `Pre-Shared Key`) are stored in encoded form in Ignition's internal gateway database. When a field is left empty, the documented fallback default is used.

> **The ADS route must already exist on the PLC.** Earlier versions registered a route as part of connecting, which is why this page used to ask for a **Route Name**, **Route Username** and **Route Password**. Those three settings are gone. Registering a route is a one-time administrative act that needs the controller's administrator credentials, and requiring them here forced those credentials into the configuration that every reading device carries for the life of the connection.
>
> Create the route on the controller before adding the device — in TwinCAT XAE, or by editing `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` directly — and the device then connects with no administrator credentials at all. A device pointed at a controller that has no matching route will fail to connect.

No TLS (default)

With **TLS Enabled** left unchecked, the driver connects over plain TCP (default port `48898`) and no encryption is applied. All other settings in the TLS Settings category are ignored. This is the default mode and requires no configuration beyond the Transport and ADS settings described above.



Secure AMS - Pre-Shared Keys

Uses TLS-PSK, where both sides share a secret key and identity string. No certificates are needed - this is the simplest way to encrypt ADS communication.

The PLC must be configured with matching PSK settings. In the TwinCAT Secure ADS configuration (usually in `/etc/TwinCAT/3.1/Target/StaticRoutes.xml`):

- Set the **Identity** to match the value in Ignition (e.g., `MYPSKUSER`)
- Set the **Password/Key** that will be used to derive the pre-shared key

```
                <Server>
                        <Tls>
                                <Psk>
                                        <Identity>MYPSKUSER</Identity>
                                        <Pwd>MySecret</Pwd>
                                </Psk>
                        </Tls>
                </Server>
        </RemoteConnections>
</TcConfig>
```

TwinCAT derives the PSK key by computing `SHA-256(Identity + Password)`. Note that the Identity must be **all uppercase**. To calculate the hex key needed for the Ignition configuration:

```
echo -n "<Identity><Password>" | shasum -a 256
```

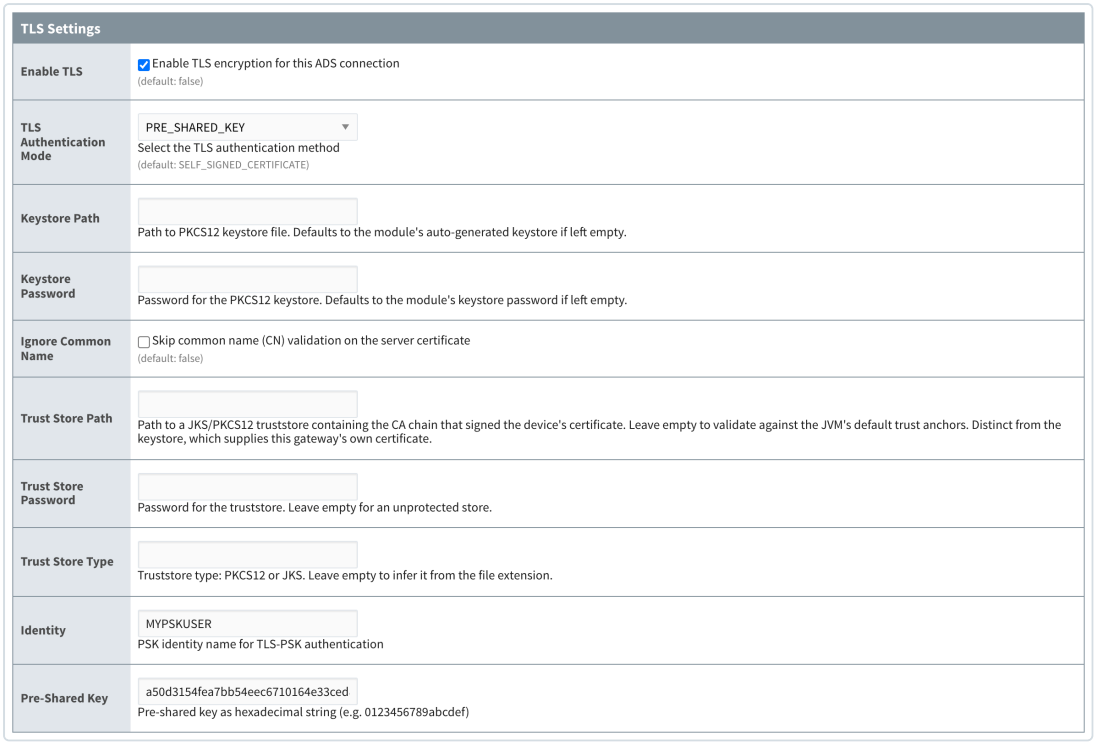
For example, with Identity `MYPSKUSER` and Password `MySecret`:

```
echo -n "MYPSKUSERMySecret" | shasum -a 256
# Result:
        a50d3154fea7bb54eec6710164e33ced872223f0a0ff05179e93e89e7f76c6f0
```

Use the resulting 64-character hex string as the **Pre-Shared Key** value in Ignition.

To enable:

1. In the **TLS Settings** category, set **TLS Enabled** to `true`
2. Set **TLS Authentication Mode** to `Pre-Shared Key`
3. Configure the PSK fields:



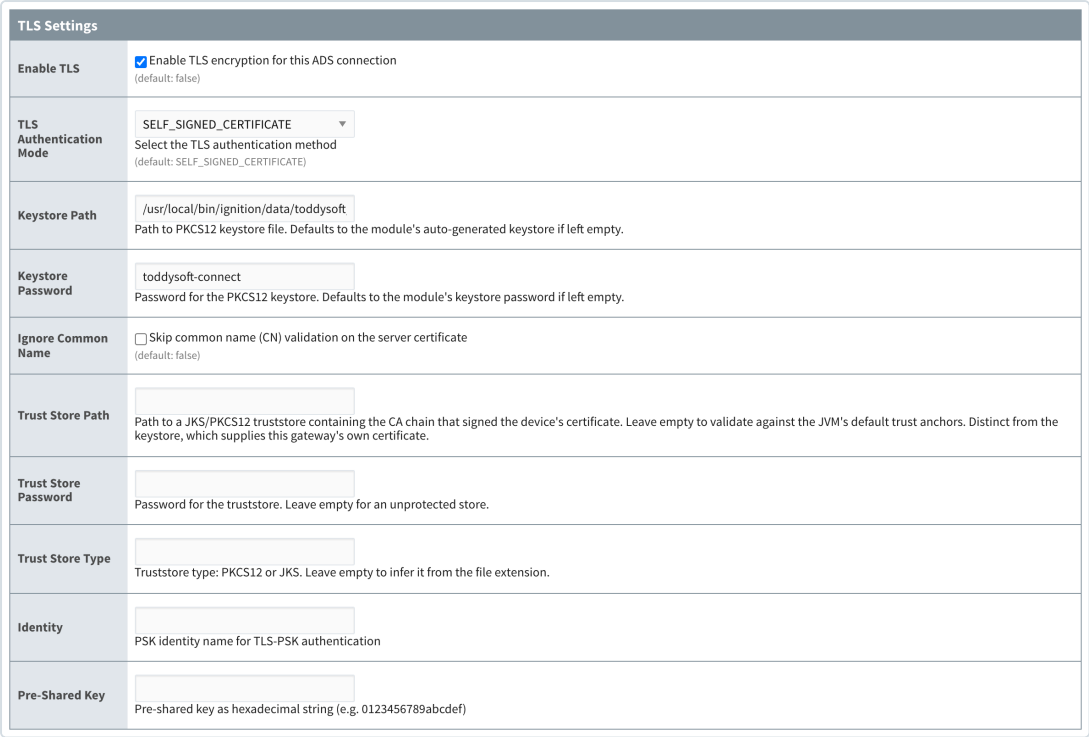
Setting	Description	Default
Identity	PSK identity string (must match the PLC configuration)	*(empty)*
Pre-Shared Key	Hex-encoded shared secret key	*(empty)*

The corresponding route entry in `/etc/TwinCAT/3.1/Target/StaticRoutes.xml` looks like this:

```
<Route>
        <Name>MyDevMachine</Name>
        <Address>192.168.24.220</Address>
        <NetId>192.168.24.220.1.1</NetId>
        <Type>TCP_IP</Type>
        <Tls>
                <Psk>
                        <Identity>MYPSKUSER</Identity>
                        <Key>a50d3154fea7bb54eec6710164e33ced872223f0a0ff05179e93e89e7f76c6f0</Key>
                </Psk>
        </Tls>
</Route>
```

Secure AMS - Self Signed Certificates

Uses TLS with automatically generated self-signed certificates. The module manages the PKCS12 keystore for you; the ADS route itself must already exist on the PLC.



To enable:

1. In the **TLS Settings** category, set **TLS Enabled** to `true`
2. Set **TLS Authentication Mode** to `Self Signed Certificate`
3. Configure the TLS-specific fields:

Setting	Description	Default
Keystore Path	Path to the PKCS12 keystore file (auto-generated if empty)	*(auto)*
Keystore Password	Password for the keystore	`toddysoft-connect`
Machine Name	Name announced to the controller in the Secure ADS session. Leave empty to announce this gateway's host name without its domain. With a self-signed certificate it must equal the certificate's common name.	*(host name)*

The settings `Identity` and `Pre-Shared Key` are ignored for this type of connection.

When a keystore path is not specified, the module automatically generates a keystore in the Ignition module settings directory.

Secure AMS - Shared Certificate Authority

Uses TLS with certificates signed by a shared Certificate Authority (CA). Both Ignition and the PLC must have certificates issued by the same CA. This is the most secure option for production environments. It uses the same TLS fields as the Self-Signed Certificate mode (shown above).

To enable:

1. In the **TLS Settings** category, set **TLS Enabled** to `true`
2. Set **TLS Authentication Mode** to `Shared Certificate Authority`
3. Configure the TLS fields:

Setting	Description	Default
Keystore Path	Path to the PKCS12 keystore containing the CA-signed certificate	*(required)*
Keystore Password	Password for the keystore	`toddysoft-connect`
Machine Name	Name announced to the controller in the Secure ADS session (see above)	*(host name)*
Trust Store Path	Path to a JKS/PKCS12 truststore holding the CA chain that signed the PLC's certificate	*(JVM defaults)*
Trust Store Password	Password for the truststore	*(none)*
Trust Store Type	`PKCS12` or `JKS`	*(from extension)*

The **Keystore** holds *this gateway's own* certificate, which it presents to the PLC. The **Trust Store** holds the CA certificate used to verify the certificate the *PLC* presents - the two are separate and serve opposite directions.

If your CA is a private one (that is, not among the public authorities the Java runtime already trusts), set **Trust Store Path** to a truststore containing it. Left empty, the driver validates the PLC certificate against the Java runtime's default trust anchors, which will reject a privately signed certificate.

Tag Addressing

Tags are addressed using symbolic names from the PLC program:

```
MAIN.bStart                    # Boolean
MAIN.iCounter                  # Integer
MAIN.fTemperature              # Float
GVL.sStatus                    # String
MAIN.stMotor.fSpeed            # Struct member
MAIN.aValues[0]                # Array element
MAIN.stArray[1].nField         # Struct array element field
```

Configure **Browse Queries** (e.g., `MAIN.*`) to control which symbols appear when browsing tags in the Ignition Designer.

Tag Groups

By default, all tags are collected in the default tag group, which means all selected tags are fetched at the default interval of once a second.

If different intervals are desired, create additional tag groups, configure the desired intervals there, and assign tags to the corresponding tag groups.

Licence diagnosis

When a device reports a licence problem, the module explains it on the gateway itself. Open **Status > Other > ToddySoft Licence (ADS)**. A device with a licence problem says so in its status and points there.

The diagnosis shows:

- **which gateway** this is, and its role if it is part of a redundant pair;
- **Ignition's licence** for this module on this gateway: activated, trial running (with its end) or trial expired;
- **what ToddySoft licence data arrived** with Ignition's licence: how many parts, whether they are complete, and a checksum that identifies the licence without revealing it;
- the **driver's own verdict** and the **cause**, in words.

Cause	Meaning	What to do
`IGNITION_UNLICENSED`	Ignition applies no activated licence to this module on this gateway (trial running or expired). ToddySoft licence data only arrives with an activated licence.	Activate a licence key that covers this module on this gateway. In a redundant pair every node needs its own activation. If a key was activated and this cause remains, see Troubleshooting: an older, expired key covering this module may be the one Ignition judges it by.
`NO_DATA_FOR_DRIVER`	The activated key carries no ToddySoft licence for this driver.	Send the report to ToddySoft.
`DATA_INCOMPLETE`	The licence data arrived incomplete or damaged.	Send the report to ToddySoft; the licence has to be delivered again.
`DATA_REJECTED`	The driver refused the delivered licence; the reason is shown.	Send the report to ToddySoft.
`RUNNING_ON_BUNDLED_TRIAL`	A licence was delivered but not accepted, and the driver runs on its built-in trial until that ends.	Send the report to ToddySoft.
`NOT_CHECKED`	Licence data arrived, but the driver gave no verdict - for instance because it could not be loaded.	Send the report to ToddySoft.
`VALID`	Licensed; tier and expiry are shown.	Nothing.

Cause	Meaning	What to do
`NOT_YET_KNOWN`	The first seconds after start-up, before Ignition reports a licence state.	Wait a moment.

Sending a report to support. The diagnosis comes as a plain-text report, the one thing to send with any licence question. The page shows it ready to select and copy. The same text is served at `http://<gateway>:<port>/data/com.toddysoft.connect.ads/licence-report` to a logged-in user. It contains no licence data, no keys, no credentials and no customer details. In a redundant pair, send the report from both nodes.

The gateway log carries the same report once each time the licence situation changes.

Troubleshooting

Licensing problems after a new or renewed licence key was activated — `DATA_INCOMPLETE` with "duplicate fragment index", or `IGNITION_UNLICENSED` with "trial expired", in the licence diagnosis:

- **Two licence keys covering this module are activated on the gateway at the same time**, typically a renewed or upgraded key activated alongside the old one. Ignition hands the module the licence data of both, in numbered parts that carry the same numbers; the module cannot tell which parts belong together, keeps one of each pair, and the licence it reassembles is unreadable. Depending on the parts it keeps, devices are refused or run on the module's bundled trial until that ends.
- If one of the two keys has **expired**, Ignition may judge the module by that key alone: the diagnosis shows `IGNITION_UNLICENSED` ("trial expired - no activated licence applies to this module") and "Data: none received", although the new key was activated successfully.
- Keep exactly **one** activated licence key that covers this module per gateway: deactivate the old one under **Config > System > Licensing**, then restart the module (or the gateway). In a redundant pair, check every node.

The module will not load, or the log shows `ClassNotFoundException` / `NoClassDefFoundError`:

- Ignition's **OPC UA** module is missing or disabled. This driver is an OPC UA device driver and cannot load without it. Install or re-enable it under **Config > System > Modules** and restart the gateway — see Installation

Device shows "Faulted":

- Verify the PLC hostname/IP is reachable
- Check that AMS Net IDs match the TwinCAT configuration
- Ensure port 48898 (or 8016 for TLS) is not blocked by a firewall
- For plain TCP, verify the PLC has an AMS route for the source AMS Net ID (see the note in ADS Settings)
- Review Gateway logs: **Status > Diagnostics > Logs**

Tags show bad quality:

- Verify the variable exists in the PLC program
- Check spelling and case sensitivity
- Ensure the PLC program is running

TLS connection fails:

- For PSK: verify the identity and key match on both sides
- For Self-Signed: check that a matching ADS route exists on the PLC
- For Shared CA: ensure both certificates are signed by the same CA

Docker related issues

When running `Ignition` inside `Docker`, the host receives an IP address on the Docker-internal network. When connecting to the PLC, the connection originates from one of the host's IP addresses. In this case the connection needs a `Source AMS Net ID` that refers to the host's IP address.

Leave the `Source AMS Port` at `0`. The driver then adopts the number the local TCP port was assigned, which is different for every connection. A fixed value works as well, but only for a single device - see Several devices on the same PLC.

VPN related issues

When connecting to a remote PLC over a VPN, depending on the type of VPN the host usually gets assigned a local VPN IP address. Similar to when using `Docker`, in this case a `Source AMS Net ID` needs to be set that references this assigned IP address.

Leave the `Source AMS Port` at `0`. The driver then adopts the number the local TCP port was assigned, which is different for every connection. A fixed value works as well, but only for a single device - see Several devices on the same PLC.

Several devices on the same PLC

The TwinCAT router routes by `Source AMS Net ID`. When two devices announce the same one to the same PLC, the router treats the second connection as a replacement for the first and closes it. The two devices then keep evicting each other, which shows up as connections that repeatedly drop and return to `Connection Lost – see the Gateway logs for details`.

Give every device that connects to the same PLC its own `Source AMS Net ID`, and add a matching route on the PLC for each of them. Leaving `Source AMS Net ID` empty does not avoid the clash: the driver then derives it from the gateway's IP address, which is the same for every device on that gateway.

The module detects the clash and reports it in the Gateway log when the device starts:

```
[Device B] Configuration conflict: source AMS Net ID
'192.168.1.100.1.1' towards 192.168.1.10
is also used by Device A. The PLC only accepts one connection per
identity and will keep closing
whichever connection it considers the older one, so these devices
will repeatedly disconnect each
other. Give each device its own identity.
```

Version history

Unreleased — 1.3.0

Built on ToddySoft Connect 1.3.0.

- **New:** licence diagnosis — why a device is or is not licensed, shown on the gateway, with a report to send to support. See Licence diagnosis.
- **New:** troubleshooting for two licence keys covering this module activated on one gateway at the same time. See Troubleshooting.
- **Fixed:** the module ships a library that every ToddySoft Connect driver needs in order to load. Without it, a device failed to start with `NoClassDefFoundError: com/toddysoft/connect/java/spi/setup/PlcDeviceSetup`.
- **Fixed:** the **Keystore Password**, **Trust Store Password** and **Pre-Shared Key** fields are masked. They were plain text boxes, so the secret was shown in clear while typing and again on every edit. Stored values are unchanged, and a device saved without touching them keeps them.
- **Fixed:** a licence could be read as corrupted when Ignition delivered it while the gateway was running. The licence check ran on one of Ignition's own threads, which cannot see the module's files, so every licence, the bundled trial included, was rejected as *License file is corrupted*, and every connection after that was refused. The check now runs as the module.
- **Fixed:** the gateway log and the audit log no longer contain the licence itself. Only its name and length are logged.
- **Fixed:** one browsed tag the driver cannot type no longer stops the whole device. Until now the device never started and kept reconnecting with `Unknown plcValueType`, without naming the tag. The tag is now skipped and named in the gateway log (search for `Skipping browsed tag`), and the device starts.

- **Fixed:** browsing a large PLC program no longer fails on the request timeout. A browse makes many round trips, so it now has at least five minutes, never less than the request timeout.
- **Changed:** the module now requires Ignition 8.1.0 instead of the development build `8.1.0-SNAPSHOT`, and its description on the gateway says it is built for Ignition 8.1, so it can be told apart from the 8.3 build once installed.
- **Changed:** date and time tags have new value types. DATE and DATE_AND_TIME are DateTime values (until now a DateTime tag holding epoch milliseconds as a number); TIME and LTIME are milliseconds, TIME_OF_DAY and LTIME_OF_DAY milliseconds since midnight (Int64). Writes take the same forms. Adjust scripts and bindings that used the old values.
- **Fixed:** a write the PLC refuses now fails in Ignition, with the reason as the status (for example `Bad_TypeMismatch` for a value of the wrong type). Until now every write was reported as successful, and the refusal appeared only in the gateway log.
- **Fixed:** a tag that has not been read yet, or whose last read failed, no longer reads 0 with Good quality. Before its first read it has no value and the quality Uncertain; after a failed read it keeps its last value with Bad quality until the next good read.
- **Fixed:** removing the last tag of a tag group (or polling rate) no longer leaves the device showing *Connection Lost*. A poll still running at that moment made the device fail with `Batch '...' not found` in the gateway log and stay in that state until it was restarted.
- **Fixed:** a tag addressed by index group and offset over a range of elements, for example `0x4040/600[0..5]:BYTE`, now reads every element; until now it read only the first. Symbolic addresses such as `MAIN.aValues` were not affected.
- **Changed:** a browsed array is one folder, named after the array and its index range (for example `arr[0..9]`), holding its elements; until now the elements sat directly beside the array's neighbours. An array of structures shows a folder per element holding its members (for example `pts[2].x`). Each element keeps its OPC item path, so existing tags keep working.

1.2.0 — 2026-09-11

Built on ToddySoft Connect 1.2.0.

- **Added:** a **Machine Name** setting for the TLS transports. Secure ADS announces a machine name to the controller, and with a self-signed certificate it must equal the certificate's common name or the controller answers `CnMismatch`. Leave it empty to announce this gateway's host name without its domain.
- **Changed:** the **Ignore Common Name** setting has been removed. It never had any effect — the parameter behind it was read by nothing, because the ignore-CN flag exists only when *registering* a route, not when connecting. Nothing changes for a device that had it ticked.
- **Fixed:** Secure ADS with the auto-generated certificate is no longer refused by every TwinCAT controller. The certificate was issued to this gateway's fully qualified

host name while Secure ADS announces the short one, so the controller compared the two and answered `CnMismatch` after a successful handshake. The certificate is now issued to the short name, and a gateway holding a stale one regenerates it at startup. **A route registered against the old certificate must be registered again.**

- **Changed:** a browsed tag now shows the name of the device it came from — `Wall-1000_MAIN.temp` rather than `MAIN.temp`, folders included. Ignition names a tag after the node it was dragged from, so without this two devices both offered `MAIN.temp` and dragging from each into one folder collided. Only the displayed name changes: the OPC item path behind every tag is untouched, so **existing tags keep working** and only newly created ones take the new name. An array element, which used to show as just `[0]`, now reads `Wall-1000_MAIN.g_arr[0]`.
- **Fixed:** a device whose PLC was switched off no longer reports **Connected**. The startup reachability check ran and its result was thrown away; with tags configured the first failed poll corrected the status within seconds, so this was only visible on a device with no tags, which polls nothing and sat claiming *Connected* indefinitely. The status is now honest from the start and the device reconnects on its own once the PLC answers. A device with no tags is also checked every five minutes so an outage is noticed at all.
- **Fixed:** the individual nodes of a browsed array read the right element again. ToddySoft Connect 1.2.0 unified how arrays are written in an address, and an array's own bounds are now part of the address the driver reports; the module was still appending the element index to that, which asks for a second array dimension rather than for an element. An array declared with a lower bound other than zero — a TwinCAT `ARRAY [1..10]` — was also read one position out. Arrays with the usual zero lower bound were unaffected.
- **Fixed:** the **TLS-SCA (Shared Certificate Authority)** mode now presents this gateway's own certificate. It differs from TLS-SSC only in how the *PLC's* certificate is verified, so it was always meant to send the keystore — but only the self-signed mode ever did. A **Keystore Path** was collected, documented as required, and then never used, so a controller configured to demand a client certificate refused the connection. No configuration change is needed; a device already filled in starts working.
- **Fixed:** connecting over **TLS-PSK** no longer destroys the pre-shared key on the controller's route. Every Secure-ADS connect sent a route *registration* message carrying the PSK identity; TwinCAT took it as one, rewrote the route from a message whose format has no field for a key, and left the route's PSK empty — so an ordinary tag read wiped out the credential it had just authenticated with. The router keeps the route in memory, so nothing looked wrong until the controller was restarted, at which point the device could no longer connect and the route had to be entered again by hand. Fixed in ToddySoft Connect 1.2.0 and confirmed on hardware. If a route's PSK has already been emptied this way, re-enter it on the controller once; it will not be cleared again.

- **Changed (action required):** the **Route Name**, **Route Username** and **Route Password** settings have been removed. ToddySoft Connect 1.2.0 no longer registers an AMS route while connecting — that is now a separate, one-time administrative operation — so those three settings bound nothing and no route was ever registered from them. For a TLS/Secure-ADS device, create the route on the controller before it connects (TwinCAT XAE, or `/etc/TwinCAT/3.1/Target/StaticRoutes.xml`); a device pointed at a controller with no matching route will fail to connect. Registering a route from inside the module is planned for a later release. Plain-TCP devices are unaffected.
- **Changed:** the keystore password and the pre-shared key are no longer written into the connection string. They are handed to the driver directly, so they no longer appear in the gateway log, in a support bundle, or in any string quoting the connection. Nothing changes on this page — the same fields, in the same places.
- **Added:** a **Limits** section with four ceilings on what the controller can make the driver allocate — browse depth, data-type table depth, and the two Sum-command sizes. The defaults are the driver's own, so an existing device connects exactly as before.

1.1.0

Built on ToddySoft Connect 1.1.0.

- **Fixed:** polling now recovers on its own after a long outage. A device whose PLC was switched off overnight, unreachable across a network outage, or idle while a licence was lapsed and later restored could keep failing every poll until the device was disabled and re-enabled by hand. The connection and tag polling are now rebuilt automatically once failures persist; existing OPC UA subscriptions are preserved.
- **Added:** three **Trust Store** settings for the *Shared Certificate Authority* mode, so a PLC certificate signed by your own CA can be validated. Previously only certificate authorities already trusted by the Java runtime could be used. Leaving **Trust Store Path** empty keeps the previous behaviour.
- **Fixed:** ADS connection checks now use the configured AMS port, so a runtime on a non-default AMS port is no longer reported as unreachable while data still flows.

1.0.0

Initial release.